

Tridib Banik

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EDUCATION

McMaster University

Sept. 2023 – Apr. 2028

Bachelor of Engineering, Software Engineering Co-op (CGPA: **3.9 / 4.0**)

Hamilton, ON

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Object-Oriented Programming, Requirements Engineering & Security, Systems Design, Concurrent Systems, Software Testing, Controls.

EXPERIENCE

Research Assistant Intern

June 2025 – Aug. 2025

CMHT Lab, McMaster Automotive Resource Centre

Hamilton, ON

- Co-developed a 2024 Ram ProMaster EV Battery Management System (BMS) manuscript, concluding that core logic uses **0.99% flash** and **9.23% RAM**, by benchmarking **7 subsystems** on an NXP automotive microcontroller.
- Designed a **Simulink** model of our team's custom **105s34p** battery pack, enabling safe and realistic verification of the complete BMS logic prior to physical deployment on the Ram ProMaster electric van.
- Accelerated onboarding for **5+** new team members by authoring **11** technical presentations totaling **308 slides** detailing software installation and hardware setup for a custom BMS.

Software Engineer Intern

Sept. 2024 – Apr. 2026

Battery Workforce EV Challenge (McMaster Team)

Hamilton, ON

- Validated **4** model-based **Simulink** subsystems, including Contactor Control, Cell Balancing, DC Fast Charging, and Fault Diagnosis, against competition specs, achieving a **95%+** unit test pass rate across **100%** of requirements.
- Shifted team testing from manual execution to automated continuous integration by architecting a **5-stage GitLab CI/CD pipeline** (verify, build, test, package, deploy) hosted on a dedicated **GitLab Runner**.
- Secured a **\$6,500 merit scholarship** for technical contributions in subsystem design, system integration, and Model-in-the-Loop (MiL) testing, supporting the team's **2nd place** finish out of **12 North American universities**.

Teaching Assistant

Jan. 2026 – Apr. 2026

Integrated Cornerstone Design Projects course, McMaster University

Hamilton, ON

- Mentored **23 students** across **5 teams** through a **6-milestone** wastewater filtration project, guiding designs from problem definition to final material selection using objective trees, regulatory compliance, and decision matrices.

PROJECTS

TrafficLightRL | Python, OpenAI Gymnasium, Stable-Baselines3, TensorFlow

- Architected a **Reinforcement Learning** pipeline in **Python** with **OpenAI Gymnasium** environments and isolated training/evaluation components, presenting it to **320+ attendees** at 2025 Canadian Undergrad Conference on AI.
- Optimized urban traffic flow against a fixed-cycle baseline by training a **PPO** agent with **Stable-Baselines3**, reducing vehicle wait times by up to **29.55%** and cutting CO₂ emissions by **35.56%** across **3** distinct traffic densities.

PrepPush | JavaScript, Chrome Extension (Manifest V3), GitHub REST API, Gemini API

- Automated interview-prep documentation for **65+ users** by publishing a **Chrome extension** that captures accepted **HackerRank** submissions and pushes them to user-owned **GitHub** repos with **Gemini**-generated Big-O analysis.
- Prevented duplicate commits on **100%** of logic-identical re-submissions by fingerprinting comment/whitespace-normalized code with **SHA-256** hashes, hardened with **429-aware** polling backoff against rate limits.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, MATLAB, HTML, CSS

Frameworks: React.js, Node.js, Flask, Django, TensorFlow, Scikit-learn, NumPy

Tools: Simulink, Simscape, Docker, Git, PostgreSQL, Linux, VS Code, Cursor, Kiro, Power Automate